

GEMINI THE TWINS

GEMINI IS A PROMINENT CONSTELLATION OF THE ZODIAC, REPRESENTING THE MYTHOLOGICAL TWINS CASTOR AND POLLUX. ITS TWO BRIGHTEST STARS MARK THE HEADS OF THE TWINS. ITEMS OF INTEREST INCLUDE A BRIGHT STAR CLUSTER AND AN UNUSUAL-LOOKING PLANETARY NEBULA.

In Greek mythology, Castor and Pollux were the sons of Queen Leda of Sparta. Pollux was said to have been fathered by the god Zeus and was immortal, while Castor's father was Leda's husband, King Tyndareus, and he was mortal. The twins joined the crew of the *Argo* and went in search of the Golden Fleece, in one of the great epics of ancient Greek mythology.

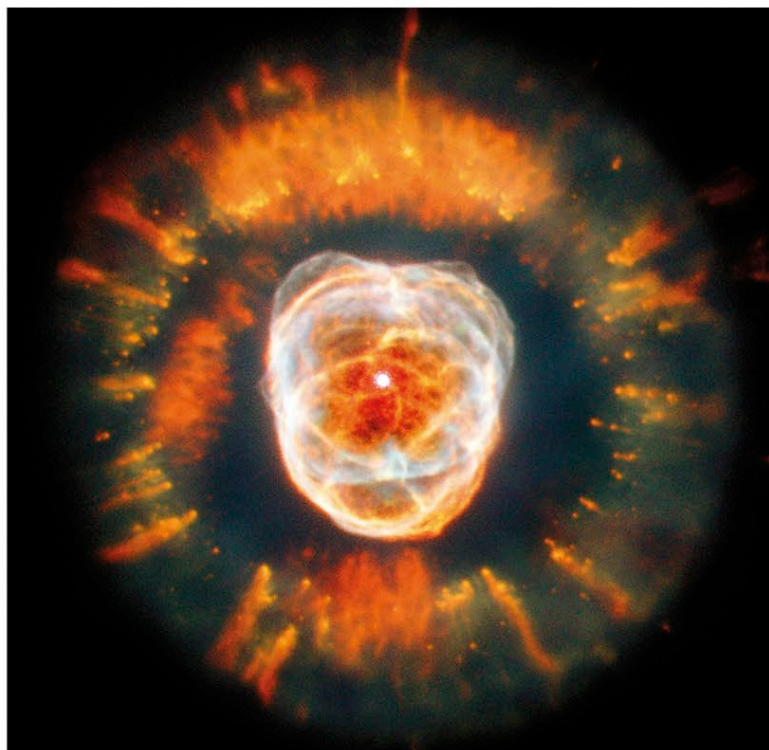
Overall, Gemini is rectangular in shape. One of its two main stars, Castor, is a remarkable multi-star system (see diagram, below right), the two brightest

members of which can be seen separately through a small telescope. Although Castor is labeled Alpha Geminorum, it is not the brightest star in Gemini, which is Pollux (Beta Geminorum), the constellation's other main star.

One of the year's richest meteor showers, the Geminids, appears to radiate from a point near Castor around December 13 each year. Up to 100 meteors an hour can be seen. Unlike most meteor showers, the parent body is not a comet but an asteroid, Phaethon.

▽ M35

This large cluster of about 200 stars is easily visible through binoculars near the border where Gemini meets Taurus. Telescopes also show the fainter and more tightly bunched cluster NGC 2158, seen at bottom left of this image. M35 is approximately 2,800 light-years away, while NGC 2158 is around 10,000 light-years more distant.



△ NGC 2392

This unusual-looking planetary nebula gets its popular name from its resemblance to a face surrounded by a furry parka hood. The "hood" is really a ring of gas streaming away from the central star, creating the appearance of a disk when seen through a small telescope. The Eskimo Nebula is about 5,000 light-years away.

KEY DATA

Size ranking 30

Brigntest stars Pollux (β)
1.1, Castor (α) 1.6

Genitive Geminorum

Abbreviation Gem

Highest in sky at 10pm
January–February

Fully visible 90°N–55°S



CHART 6

MAIN STARS

Castor Alpha (α) Geminorum
Blue-white multiple star

1.6 ↔ 51 light-years

Pollux Beta (β) Geminorum
Orange giant

1.1 ↔ 34 light-years

Alhena Gamma (γ) Geminorum
Blue-white subgiant

1.9 ↔ 110 light-years

Wasat Delta (δ) Geminorum
White main-sequence star

3.5 ↔ 60 light-years

Mebsuta Epsilon (ε) Geminorum
Yellow supergiant

3.0 ↔ 845 light-years

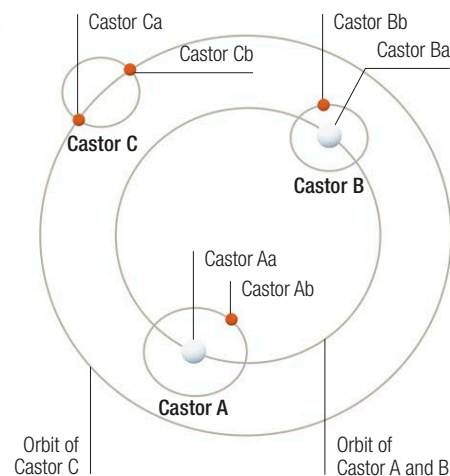
DEEP-SKY OBJECTS

M35

Large, bright open cluster of about 200 stars

NGC 2392 (Eskimo Nebula)

Planetary nebula, also called the Clown Face Nebula



△ Castor multi-star system

Through a small telescope, Castor appears as a double star, Castor A and B, with components that orbit one another every 460 years. Each of these stars is itself a binary. To complicate the picture further, Castor A and B also have a faint red dwarf companion, known as Castor C, which is an eclipsing binary, completing a remarkable system of six stars all linked by gravity.

Stars labeled Alpha (α) are not always the brightest in their constellations—an example is **Castor**, which is **dimmer** than **Pollux**